

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

17%

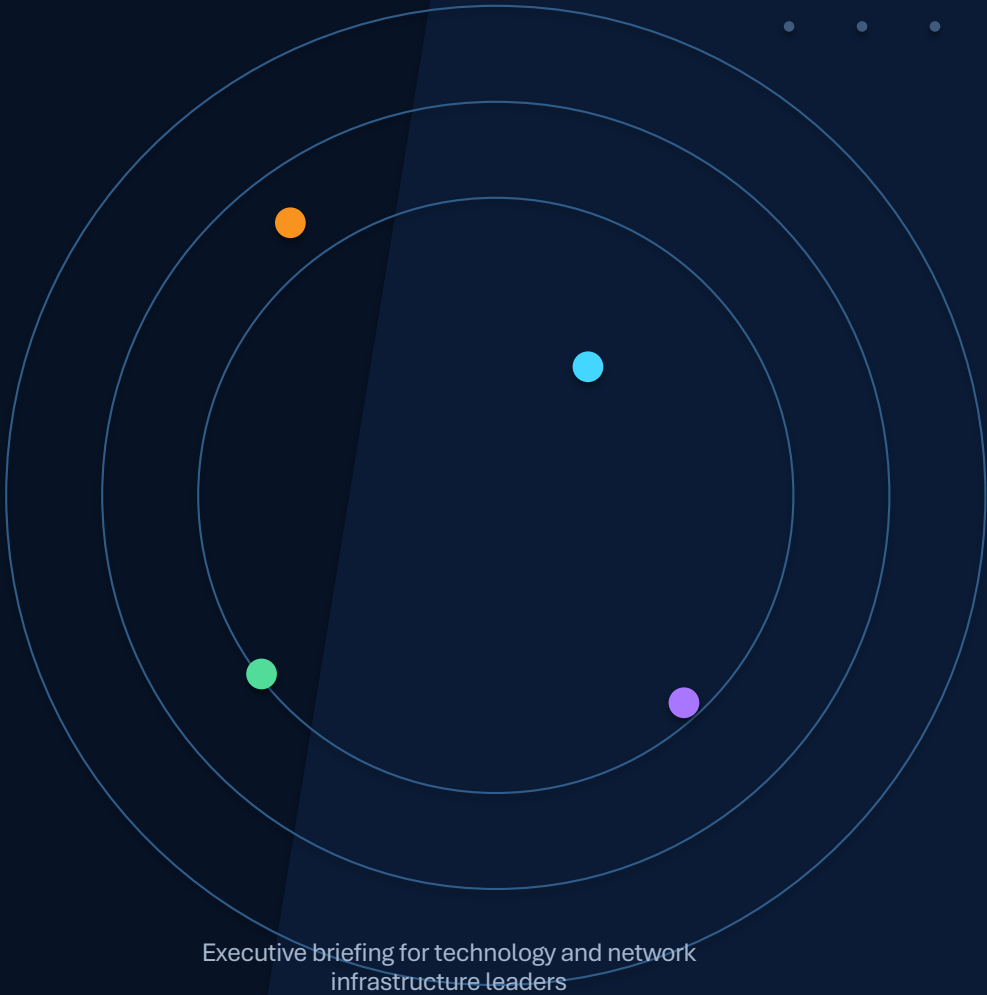
Global traffic growth

225

Major disruptions

6.5%

Traffic mitigated



17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Internet traffic grew year over year

Normalized after post-holiday activity

3.3x

Starlink traffic surge in 2024

Satellite connectivity expanded to underserved locations

41%

Global traffic from mobile devices



Starlink trend line is scaled relative to peak observed request volume; service was not globally available.

Google Leads Top 10 Services as Generative AI Rises

Top 10 most popular Internet services

- #1 Google
- #2 Facebook
- #3 Apple
- #4 TikTok
- #5 AWS
- #6 Microsoft
- #7 Instagram
- #8 YouTube
- #9 Amazon
- #10 WhatsApp

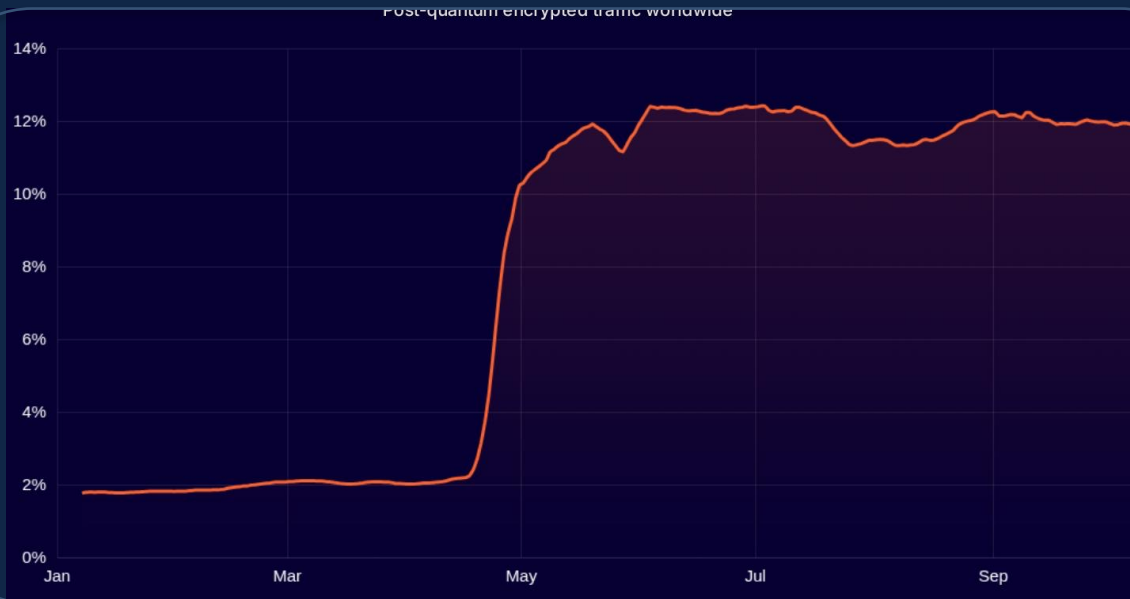
Methodology: aggregate data from Cloudflare 1.1.1.1 DNS resolver.

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

AI signal

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



Chrome 124 enabled post-quantum key agreement by default on April 17; Firefox rollout and Safari testing followed.

- **Mobile OS:** 33% of traffic came from iOS devices; Android and iOS remain dominant.
- **Search referrals:** Google accounted for 89%, followed by Yandex, Baidu, Bing and DuckDuckGo.

2%

Early 2024 TLS
1.3 share

13%

Approx.
adoption peak

21%

HTTP/3
traffic

API client language leaders

Go		~12%
Python		9.6%
Java		7.4%
.NET		3.6%

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

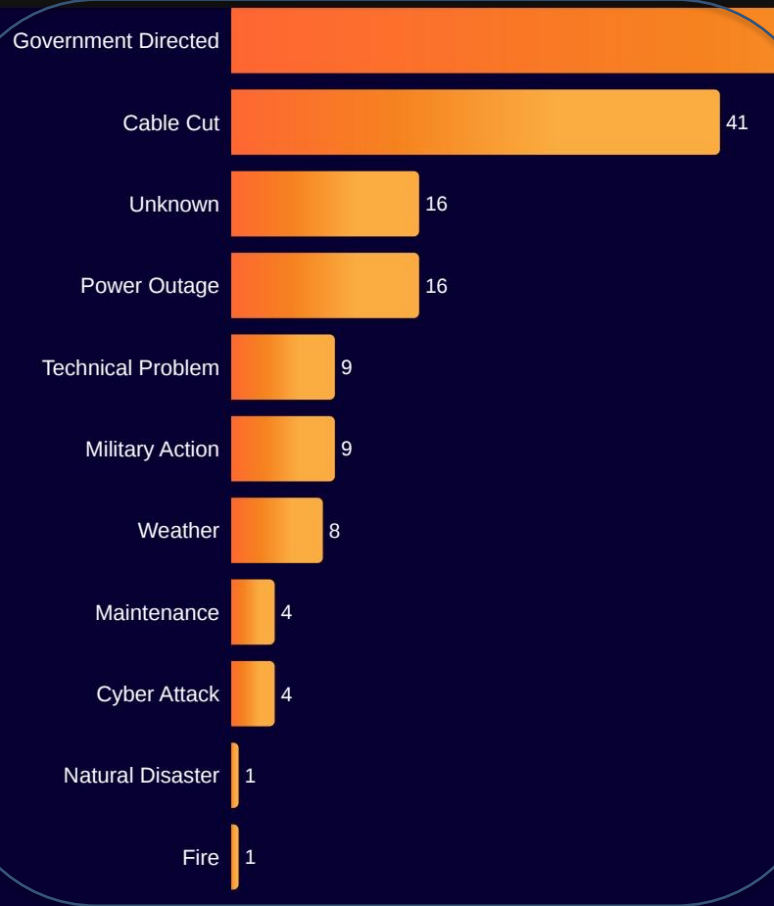
Major Internet
disruptions
observed globally

116+

Government-
directed
shutdowns

41

Cable cut
incidents



Cause breakdown from the brief

Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

IPv6 reached 28% of dual-stack traffic in 2024.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration



6.5%

All processed traffic mitigated

Threat scale across Cloudflare network

68%

Bot traffic from top 10 countries

Concentrated sources

4.3%

Emails processed were malicious

43%

Malicious emails with deceptive link

Most-attacked sector

Gambling and Games

Cloudflare identified Gambling/Games as the most attacked industry category in 2024.

- **Log4j persists:** CVE-2021-44228 remains actively targeted more than two years after disclosure.
- **Routing security:** RPKI valid routes increased by 6.4%.
- **Email origin:** .bar generated the largest share of malicious and spam email observed.

Executive implication: mitigation, patch governance, bot visibility and email controls remain board-level risk priorities.

Five Takeaways Shaping the 2025 Internet Outlook



1

Capacity planning must assume continued growth

Global traffic rose 17%, while Starlink grew 3.3x as satellite connectivity expanded.

2

Cryptographic modernization accelerated

Post-quantum encryption jumped from ~2% to ~13% of TLS 1.3 traffic after Chrome 124.

3

Internet resilience is geopolitical

Government-directed shutdowns led 225 disruptions, ahead of cable cuts and power outages.

4

Threat exposure remains persistent and concentrated

6.5% of traffic was mitigated; Log4j exploitation and bot concentration continue.

5

Developer ecosystems are shifting

Go became the top API client language with roughly 12% of automated API requests.

2025 planning lens: build for higher demand, faster protocol change, policy-driven outages, and persistent adversary pressure.